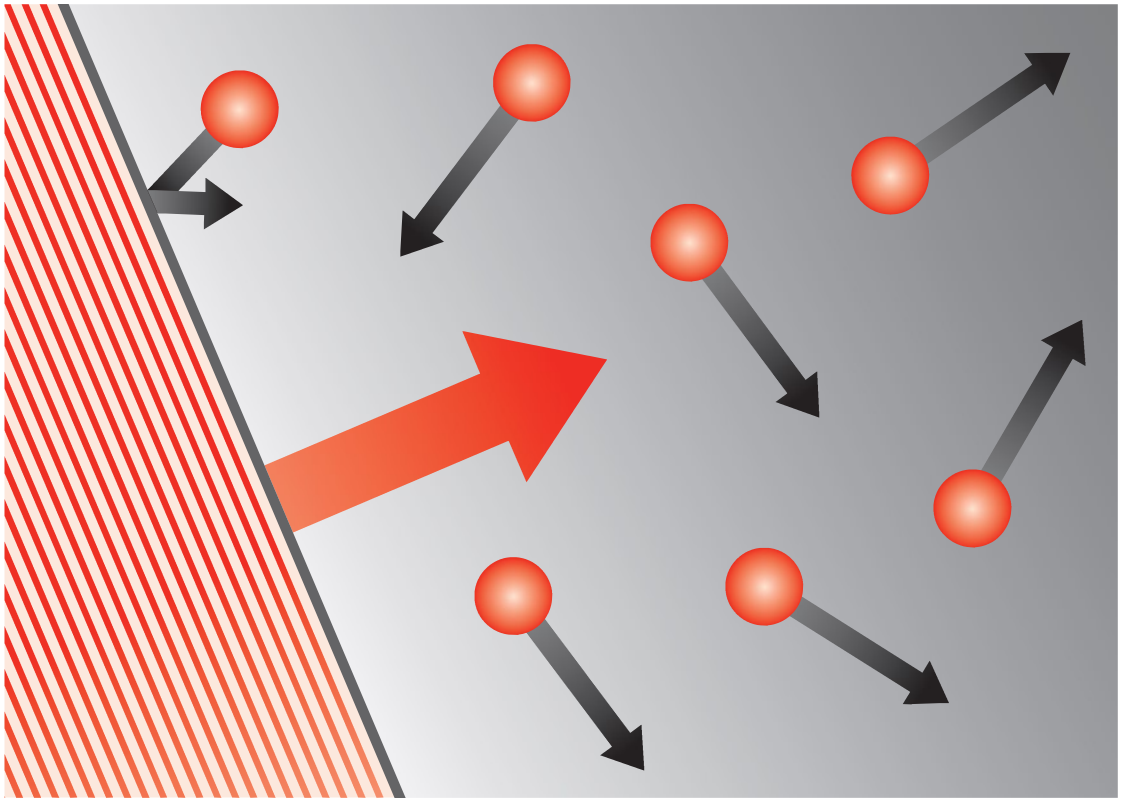
CHAPTER 2

# Introduction to Conduction



Recall that *conduction* is the transport of energy in a medium due to a temperature gradient, and the physical mechanism is one of random atomic or molecular activity. In Chapter 1 we learned that conduction heat transfer is governed by *Fourier's law* and that use of the law to determine the heat flux depends on knowledge of the manner in which temperature varies within the medium (the *temperature distribution*). By way of introduction, we restricted our attention to simplified conditions (one-dimensional, steady-state conduction in a plane wall). However, Fourier's law is applicable to transient, multidimensional conduction in complex geometries.

The objectives of this chapter are twofold. First, we wish to develop a deeper understanding of Fourier's law. What are its origins? What form does it take for different geometries? How does its proportionality constant (the *thermal conductivity*) depend on the physical nature of the medium? Our second objective is to develop, from basic principles, the general equation, termed the *heat equation*, which governs the temperature distribution in a medium. The solution to this equation provides knowledge of the temperature distribution, which may then be used with Fourier's law to determine the heat flux.

## 2.1 The Conduction Rate Equation

Although the conduction rate equation, Fourier's law, was introduced in Section 1.2, it is now appropriate to consider its origin. Fourier's law is *phenomenological*; that is, it is developed from observed phenomena rather than being derived from first principles. Hence, we view the rate equation as a generalization based on much experimental evidence. For example, consider the steady-state conduction experiment of Figure 2.1. A cylindrical rod of known material is insulated on its lateral surface, while its end faces are maintained at different temperatures, with $T_1 > T_2$. The temperature difference causes conduction heat transfer in the positive $x$-direction. We are able to measure the heat transfer rate $q_x$, and we seek to determine how $q_x$ depends on the following variables: $\Delta T$, the temperature difference; $\Delta x$, the rod length; and $A$, the cross-sectional area.

We might imagine first holding $\Delta T$ and $\Delta x$ constant and varying $A$. If we do so, we find that $q_x$ is directly proportional to $A$. Similarly, holding $\Delta T$ and $A$ constant, we observe that $q_x$ varies inversely with $\Delta x$. Finally, holding $A$ and $\Delta x$ constant, we find that $q_x$ is directly proportional to $\Delta T$. The collective effect is then

$$q_x \propto A \frac{\Delta T}{\Delta x}$$

In changing the material (e.g., from a metal to a plastic), we would find that this proportionality remains valid. However, we would also find that, for equal values of $A$, $\Delta x$, and $\Delta T$,

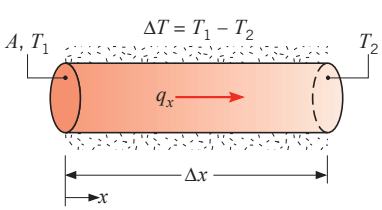



**FIGURE 2.1** Steady-state heat conduction experiment.

the value of $q_x$ would be smaller for the plastic than for the metal. This suggests that the proportionality may be converted to an equality by introducing a coefficient that is a measure of the material behavior. Hence, we write

$$q_x = kA\frac{\Delta T}{\Delta x}$$

where $k$, the *thermal conductivity* (W/m·K) is an important *property* of the material. Evaluating this expression in the limit as $\Delta x \to 0$, we obtain for the heat *rate*

$$q_x = -kA\frac{dT}{dx} \tag{2.1}$$

or for the heat *flu*

$$q''_x = \frac{q_x}{A} = -k\frac{dT}{dx} \tag{2.2}$$

Recall that the minus sign is necessary because heat is always transferred in the direction of decreasing temperature.

Fourier's law, as written in Equation 2.2, implies that the heat flux is a directional quantity. In particular, the direction of $q''_x$ is *normal* to the cross-sectional area $A$. Or, more generally, the direction of heat flow will always be normal to a surface of constant temperature, called an *isothermal* surface. Figure 2.2 illustrates the direction of heat flow $q''_x$ in a plane wall for which the *temperature gradient* $dT/dx$ is negative. From Equation 2.2, it follows that $q''_x$ is positive. Note that the isothermal surfaces are planes normal to the $x$-direction.

Recognizing that the heat flux is a vector quantity, we can write a more general statement of the conduction rate equation (*Fourier's law* ) as follows:

$$\boldsymbol{q}'' = -k\nabla T = -k\left(\boldsymbol{i}\frac{\partial T}{\partial x} + \boldsymbol{j}\frac{\partial T}{\partial y} + \boldsymbol{k}\frac{\partial T}{\partial z}\right) \tag{2.3}$$

where $\nabla$ is the three-dimensional del operator and $T(x, y, z)$ is the scalar temperature field. It is implicit in Equation 2.3 that the heat flux vector is in a direction perpendicular to the isothermal surfaces. An alternative form of Fourier's law is therefore

$$\boldsymbol{q}'' = q''_n\boldsymbol{n} = -k\frac{\partial T}{\partial n}\boldsymbol{n} \tag{2.4}$$

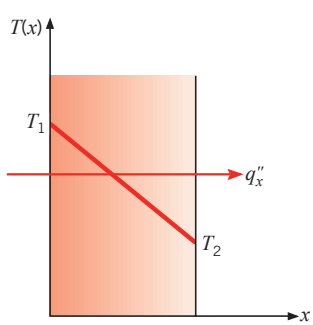



**FIGURE 2.2** The relationship between coordinate system, heat flow direction, and temperature gradient in one dimension.

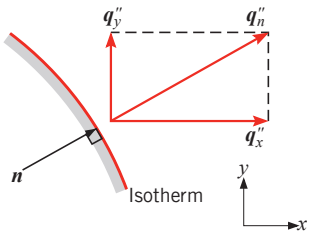



**FIGURE 2.3** The heat flux vector normal to an isotherm in a two-dimensional coordinate system.

where $q''_n$ is the heat flux in a direction $n$, which is normal to an *isotherm*, and $\boldsymbol{n}$ is the unit normal vector in that direction. This is illustrated for the two-dimensional case in Figure 2.3. The heat transfer is sustained by a temperature gradient along $\boldsymbol{n}$. Note also that the heat flux vector can be resolved into components such that, in Cartesian coordinates, the general expression for $\boldsymbol{q}''$ is

$$\boldsymbol{q}'' = \boldsymbol{i}q''_x + \boldsymbol{j}q''_y + \boldsymbol{k}q''_z \tag{2.5}$$

where, from Equation 2.3, it follows that

$$q''_x = -k\frac{\partial T}{\partial x} \qquad q''_y = -k\frac{\partial T}{\partial y} \qquad q''_z = -k\frac{\partial T}{\partial z} \tag{2.6}$$

Each of these expressions relates the heat flux *across a surface* to the temperature gradient in a direction perpendicular to the surface. It is also implicit in Equation 2.3 that the medium in which the conduction occurs is *isotropic.* For such a medium, the value of the thermal conductivity is independent of the coordinate direction.

Fourier's law is the cornerstone of conduction heat transfer, and its key features are summarized as follows. It is *not* an expression that may be derived from first principles; it is instead a generalization based on experimental evidence. It is an expression that *define* an important material property, the thermal conductivity. In addition, Fourier's law is a vector expression indicating that the heat flux is normal to an isotherm and in the direction of decreasing temperature. Finally, note that Fourier's law applies for all matter, regardless of its state (solid, liquid, or gas).

## 2.2 The Thermal Properties of Matter

To use Fourier's law, the thermal conductivity of the material must be known. This property, which is referred to as a *transport property,* provides an indication of the rate at which energy is transferred by the diffusion process. It depends on the physical structure of matter, atomic and molecular, which is related to the state of the matter. In this section we consider various forms of matter, identifying important aspects of their behavior and presenting typical property values.

### 2.2.1 Thermal Conductivity

From Fourier's law, Equation 2.6, the thermal conductivity associated with conduction in the $x$-direction is defined as

$$k_x \equiv -\frac{q''_x}{(\partial T/\partial x)}$$

Similar definitions are associated with thermal conductivities in the $y$- and $z$-directions ($k_y$, $k_z$), but for an isotropic material the thermal conductivity is independent of the direction of transfer, $k_x = k_y = k_z \equiv k$.

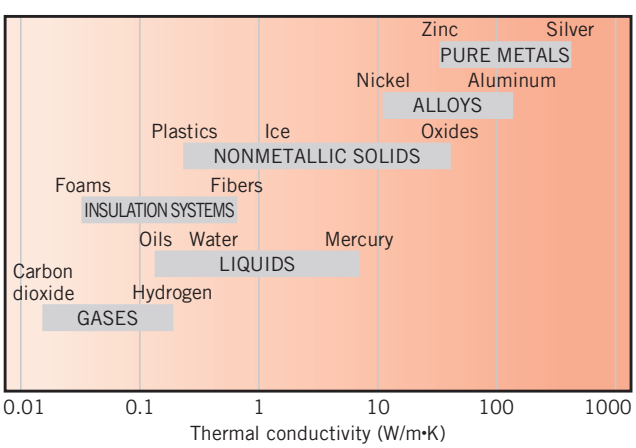



**FIGURE 2.4** Range of thermal conductivity for various states of matter at normal temperatures and pressure.